



2011 Half-Yearly Examination

FORM VI

MATHEMATICS 2 UNIT

Monday 28th February 2011

General Instructions

- Writing time — 2 hours
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.
- All necessary working should be shown in every question.
- Start each question in a new booklet.

Structure of the paper

- Total marks — 96
- All six questions may be attempted.
- All six questions are of equal value.

Collection

- Write your candidate number clearly on each booklet.
- Hand in the six questions in a single well-ordered pile.
- Hand in a booklet for each question, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place the question paper inside your answer booklet for Question One.

Checklist

- SGS booklets — 6 per boy
- Candidature — 87 boys

Examiner

FMW

QUESTION ONE (16 marks) Use a separate writing booklet.

Marks

(a) Differentiate:

(i) $3x^4$ 1

(ii) $(x - 6)^5$ 1

(iii) e^{3x-1} 1

(b) Find a primitive of:

(i) $2x^3 - 3$ 1

(ii) $2e^{2x}$ 1

(c) Consider the parabola with equation $x^2 = 8y$.

(i) Write down the coordinates of its focus. 1

(ii) What is the equation of its directrix? 1

(d) Evaluate $\int_1^4 x \, dx$. 2

(e) Write $\frac{2}{e}$ correct to 2 decimal places. 1

(f) Consider the curve whose gradient function is $y' = (x - 2)(x - 3)(x + 4)$. For what values of x is the curve stationary? 1

(g) Consider the curve whose concavity function is $y'' = 3x - 2$. For what values of x is the curve concave down? 1

(h) Write down the centre and radius of the circle with equation $(x - 2)^2 + (y + 3)^2 = 9$. 2

(i) Find the gradient of the tangent to the curve $y = 2\sqrt{x}$ at the point $(9, 6)$. 2

QUESTION TWO (16 marks) Use a separate writing booklet.**Marks**

- (a) Sketch the graph of $y = e^x + 1$, showing any intercepts with the x or y axes and any asymptotes. **2**

(b)

x	2	3	4
$f(x)$	7	5	3

2

Use Simpson's rule with the 3 function values in the table above to approximate $\int_2^4 f(x) dx$.

- (c) Find:

(i) $\int e^{-3x+2} dx$ **1**

(ii) $\int x(x^2 - 2) dx$ **1**

(iii) $\int \frac{x^3 + 2x}{x} dx$ **1**

- (d) A curve has gradient function $\frac{dy}{dx} = 4x - 2$ and passes through the point $(3, 10)$. Find the equation of the curve. **2**

- (e) Differentiate:

(i) $y = (5x - 2)^7$ **1**

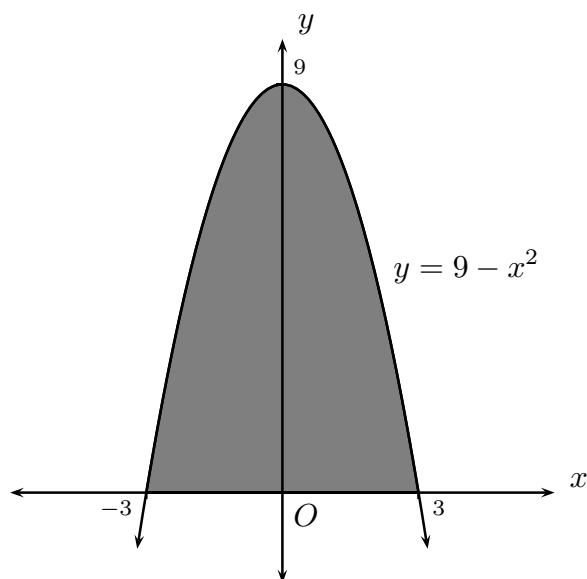
(ii) $y = \frac{x}{e^x}$ **2**

Question Two Continues On the Next Page

QUESTION TWO (Continued)

(f) (i)

3



Calculate the area of the shaded region in the diagram above.

(ii) Hence write down the value of $\int_0^9 \sqrt{9-y} \, dy$.

1

QUESTION THREE (16 marks) Use a separate writing booklet.

Marks

(a) Find the equation of the tangent to the curve $y = \frac{2}{x}$ at the point where $x = 2$.

3

(b) (i) Differentiate $f(x) = (4x^3 - 5)^5$.

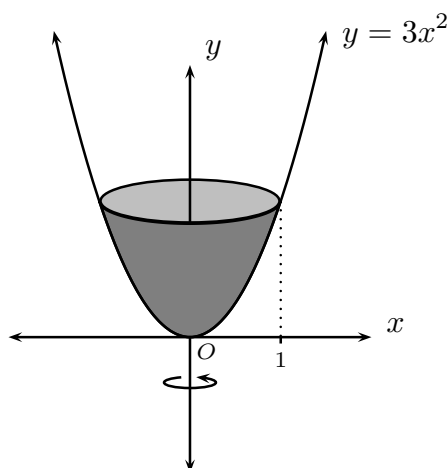
1

(ii) Hence find $\int x^2(4x^3 - 5)^4 dx$.

1

(c)

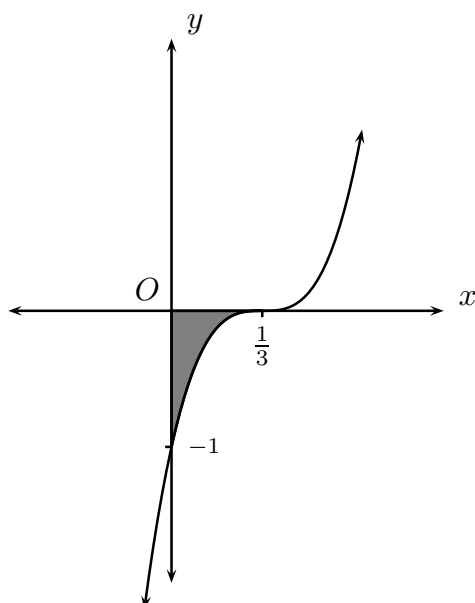
3



As shown in the diagram above, the part of the curve $y = 3x^2$ between $x = 0$ and $x = 1$ is rotated about the y -axis to form a cup. Show that the volume of the cup is $\frac{3\pi}{2}$ cubic units.

(d)

3



The diagram above shows the curve $y = (3x - 1)^3$. Find the area of the shaded region.

Question Three Continues On the Next Page

Exam continues overleaf ...

QUESTION THREE (Continued)

(e) Find the equations of the parabolas with:

(i) vertex $(0, 0)$ and focus $(1, 0)$,

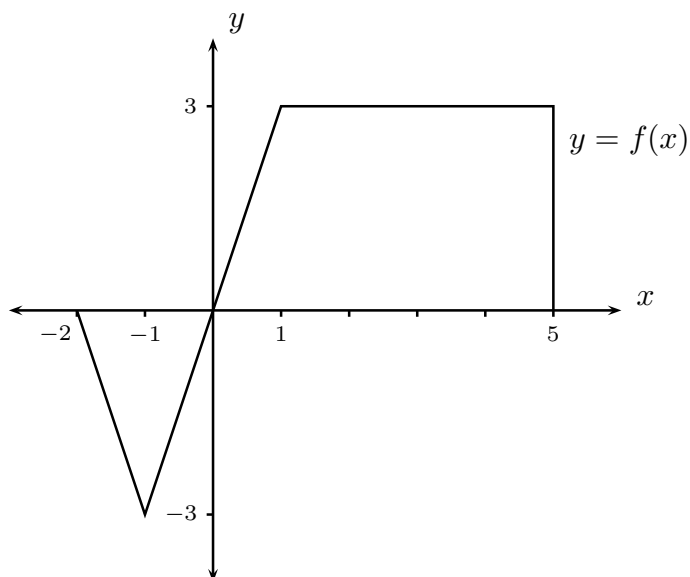
(ii) focus $(1, 3)$ and directrix $y = -1$.

1

2

(f)

2



Given the sketch of $y = f(x)$ drawn above, use area formulae to find $\int_{-2}^5 f(x) dx$.

QUESTION FOUR (16 marks) Use a separate writing booklet.

Marks

(a) Consider the curve with equation $y = x^3 - 5x^2 + 7x$.

(i) Show that $y' = (3x - 7)(x - 1)$ and find y'' .

2

(ii) Find the two stationary points and determine their nature.

3

(iii) Find any points of inflexion.

2

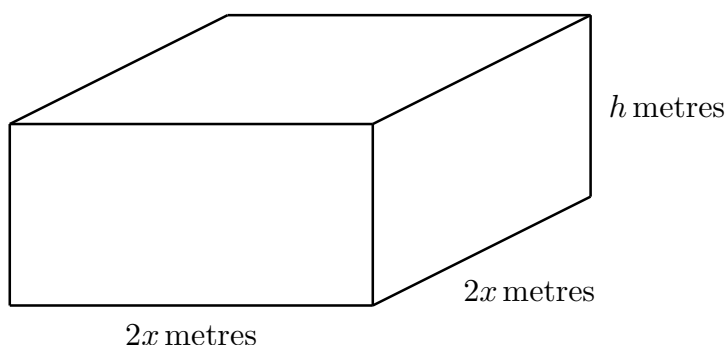
(iv) Sketch the curve using the above information.

2

(v) What is the maximum value of $y = x^3 - 5x^2 + 7x$ in the interval from $0 \leq x \leq 5$?

1

(b)



A closed box with a square base is made from a piece of cardboard, as shown in the diagram above. The area of cardboard used is 6 square metres. Let h metres be the height of the box and let $2x$ metres be the side length of the base.

(i) Show that $h = \frac{3}{4x} - x$.

2

(ii) Hence show that the volume, V cubic metres, of the box is given by the formula $V = 3x - 4x^3$.

1

(iii) Use calculus to find the maximum volume of the box.

3

QUESTION FIVE (16 marks) Use a separate writing booklet.**Marks**

- (a) (i) Express the equation $y^2 + 6y + 4x - 3 = 0$ in the form $(y - k)^2 = -4a(x - h)$.

2

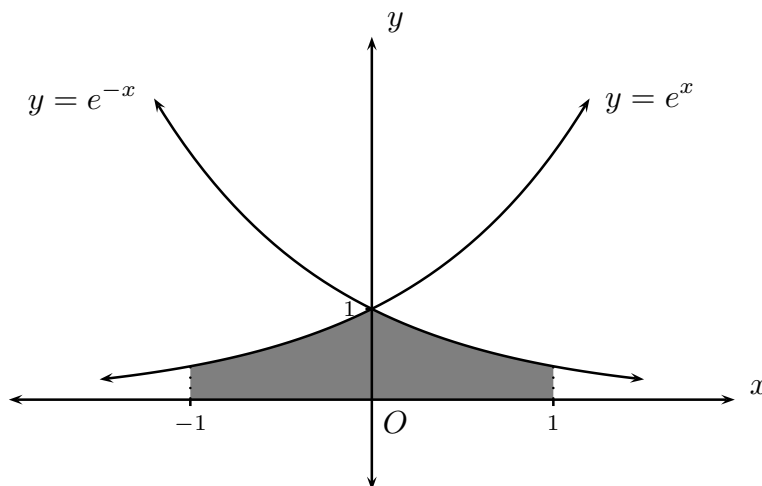
- (ii) Hence find the coordinates of the vertex and the focus of the parabola $y^2 + 6y + 4x - 3 = 0$.

2

- (b) Find $\int_0^3 (2 - \frac{1}{3}x)^{-3} dx$.

3

- (c)

2

Calculate the exact area of the shaded region in the diagram above.

- (d) (i) Sketch the region bounded by the curves $y = e^{2x}$, $y = e^{-x}$ and the line $x = 1$.

1

- (ii) If this region is rotated about the x -axis, find the exact volume of the solid formed.

3

- (e) Find the value of k given that $\int_k^0 \frac{1}{e^x} dx = e^2 - 1$.

3

QUESTION SIX (16 marks) Use a separate writing booklet.

Marks

- (a) (i) Copy and complete the following table using exact values:

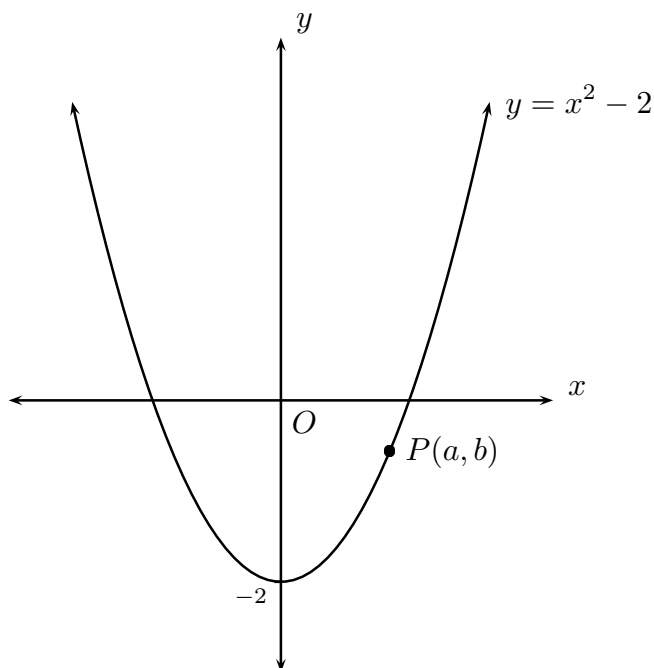
1

x	-1	0	1
$f(x) = \frac{1}{1+e^{-x}}$			

- (ii) Use the trapezoidal rule with the 3 function values from your table to find the value of $\int_{-1}^1 \frac{1}{1+e^{-x}} dx$.

3

(b)



The diagram above shows the curve $y = x^2 - 2$ and the point $P(a, b)$ on the curve.

- (i) Find the equation of the normal at P .
- (ii) Find all possible points P on the curve such that the normal at P passes through $(0, 0)$.

2

3

Question Six Continues On the Next Page

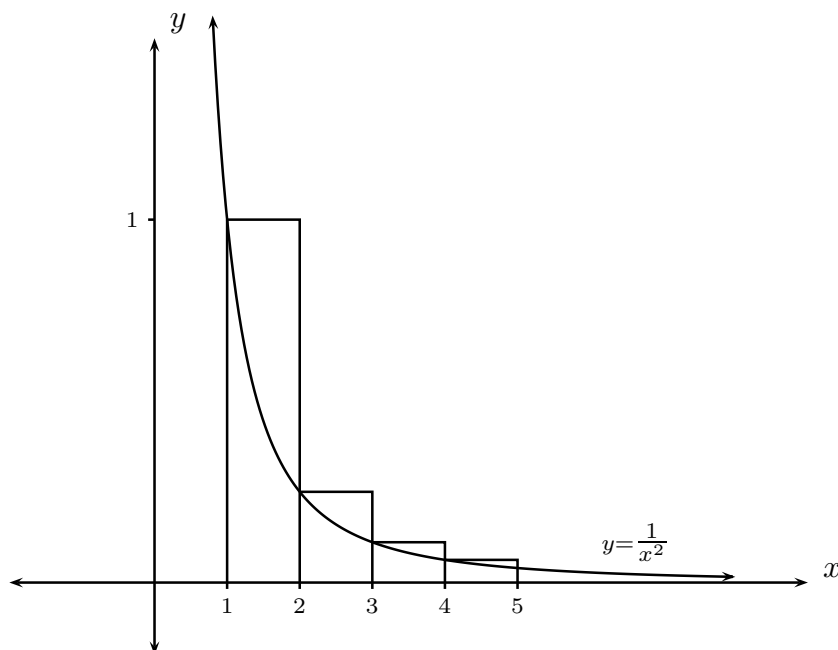
QUESTION SIX (Continued)

(c) (i) Evaluate $\int_1^5 \frac{1}{x^2} dx$.

2

(ii)

2



The diagram above shows part of the curve $y = \frac{1}{x^2}$. Rectangles of width 1 unit are constructed as shown. Use the rectangles in the diagram and part (i) to explain why

$$1 \times \frac{1}{1^2} + 1 \times \frac{1}{2^2} + 1 \times \frac{1}{3^2} + 1 \times \frac{1}{4^2} > \frac{4}{5}.$$

(iii) Show that

3

$$1 + \frac{1}{4} + \frac{1}{9} + \cdots + \frac{1}{10\,000} > \frac{100}{101}.$$

END OF EXAMINATION

B L A N K P A G E

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Q.1

(a) (i) $\frac{d}{dx}(3x^4) = 12x^3$ ✓

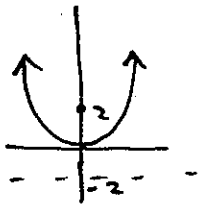
(ii) $\frac{d}{dx}((x-6)^5) = 5(x-6)^4$ ✓

(iii) $\frac{d}{dx}(e^{3x-1}) = 3e^{3x-1}$ ✓

(b) (i) $\frac{2x^4}{4} - 3x = \frac{1}{2}x^4 - 3x + c$ ✓

(ii) $e^{2x} + c$ ✓

(c) $x^2 = 8y$
 $= 4(2)y$ (i) focus $(0, 2)$ ✓
(ii) directrix $y = -2$ ✓



(d) $\int_1^4 x dx = \left[\frac{x^2}{2} \right]_1^4$ ✓
 $= \frac{16}{2} - \frac{1}{2}$
 $= \frac{15}{2}$ ✓

(e) $\frac{2}{e} \approx 0.74$ (2 d.p.) ✓

(f) $y' = 0$ at $x = 2, 3, -4$ ✓

(g) $3x - 2 < 0$
 $3x < 2$
 $x < \frac{2}{3}$ ✓

(h) centre $(2, -3)$, radius 3 ✓

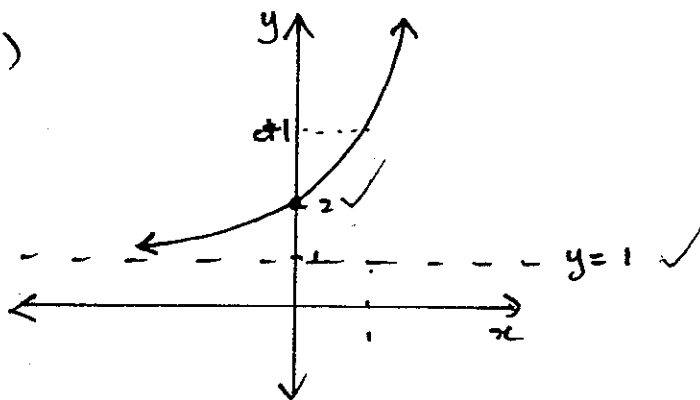
(i) $y = 2x^{\frac{1}{2}}$
 $y' = 2 \times \frac{1}{2} x^{-\frac{1}{2}}$ ✓ at $x = 9$
 $y' = 9^{-\frac{1}{2}} = \frac{1}{3}$ ✓

do not
penalise
omission of
constants

/ 16

Q2

(a)



$$(b) \int_2^4 f(x) dx \div 1 \times \frac{1}{3} [7 + 4(5) + 3] \checkmark$$

$$= 10 \checkmark$$

$$(c) (i) \int e^{-3x+2} dx = -\frac{1}{3} e^{-3x+2} + C \checkmark$$

$$(ii) \int x(x^2-2) dx = \int x^3 - 2x dx$$

$$= \frac{x^4}{4} - x^2 + C \checkmark$$

$$(iii) \int \frac{x^3+2x}{x} dx = \int x^2 + 2 dx$$

$$= \frac{x^3}{3} + 2x + C \checkmark$$

(do not penalise omission of constants!)

$$(d) \frac{dy}{dx} = 4x - 2$$

$$y = 2x^2 - 2x + C \checkmark$$

$$10 = 2(3)^2 - 2(3) + C$$

$$10 = 12 + C$$

$$C = -2 \checkmark$$

$$y = 2x^2 - 2x - 2$$

$$(e) (i) y = (5x-2)^7$$

$$y' = 7(5x-2)^6 \times 5 \checkmark$$

$$= 35(5x-2)^6 \checkmark$$

$$(ii) y = \frac{x}{e^x}$$

$$y' = \frac{e^x \times 1 - x \times e^x}{(e^x)^2} \checkmark$$

$$= \frac{e^x (1-x)}{e^{2x}}$$

$$= \frac{1-x}{e^x} \checkmark$$

$$(f) (i) A = 2 \int_0^3 (9-x^2) dx \checkmark$$

$$= 2 \left[9x - \frac{x^3}{3} \right]_0^3 \checkmark$$

$$= 2(27 - \frac{27}{3} - 0)$$

$$= 36 \text{ square units} \checkmark$$

$$(ii) \int_0^9 \sqrt{9-y} dy = \frac{1}{2} \times 36$$

$$= 18 \checkmark$$

(note: $x^2 = 9-y$
 $x = \sqrt{9-y}$ -
 RHS of curve given)

Q 3

$$(a) \quad y = \frac{2}{x}$$

$$= 2x^{-1}$$

$$y' = -2x^{-2} \quad \checkmark$$

$$= -\frac{2}{x^2}$$

$$\text{at } x=2, \quad y' = -\frac{2}{2^2}$$

$$y = 1$$

$$= -\frac{1}{2}$$

tangent :

$$y - 1 = -\frac{1}{2}(x - 2) \quad \checkmark \checkmark$$

$$2y - 2 = -x + 2$$

$$x + 2y - 4 = 0$$

$$(b) \quad (i) \quad f(x) = (4x^3 - 5)^5$$

$$f'(x) = 5(4x^3 - 5)^4 \times 12x^2$$

$$= 60x^2(4x^3 - 5)^4 \quad \checkmark$$

$$(ii) \quad \int x^2(4x^3 - 5)^4 dx$$

$$= \frac{1}{60} \int 60x^2(4x^3 - 5)^4 dx$$

$$= \frac{1}{60} (4x^3 - 5)^5 + C \quad \checkmark$$

$$(c) \quad y = 3x^2$$

$$\frac{y}{3} = x^2$$

$$V = \pi \int_0^3 \frac{y}{3} dy \quad \checkmark \checkmark$$

$$= \pi \left[\frac{y^2}{6} \right]_0^3$$

$$= \pi \left(\frac{9}{6} - 0 \right) \quad \checkmark$$

$$= \frac{3\pi}{2} \quad \text{as required}$$

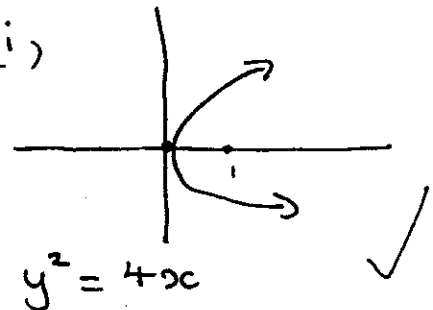
$$(d) \quad A = - \int_0^{\frac{1}{3}} (3x-1)^3 dx \quad \checkmark$$

$$= - \left[\frac{(3x-1)^4}{4 \times 3} \right]_0^{\frac{1}{3}} \quad \checkmark$$

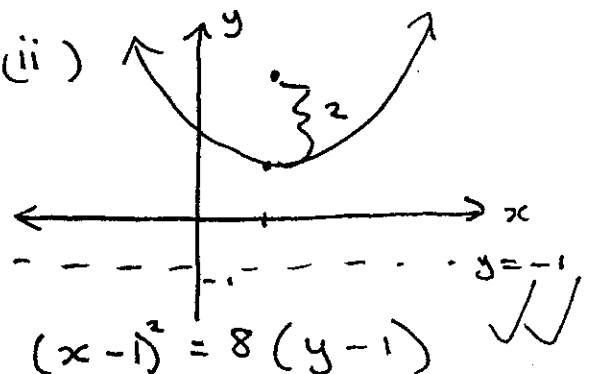
$$= - \left(0 - \frac{1}{12} \right)$$

$$= \frac{1}{12} \text{ square unit} \quad \checkmark$$

(e) (i)



(ii)



$$(f) \quad \int_{-2}^5 f(x) dx = -\frac{1}{2} \times 2 \times 3 + \frac{1}{2} \times 3(5+4) \quad \checkmark$$

$$= -3 + \frac{27}{2}$$

$$= 10\frac{1}{2} \quad \checkmark$$

16

Q4

(a) $y = x^3 - 5x^2 + 7x$

(i) $y' = 3x^2 - 10x + 7$
 $= 3x^2 - 3x - 7x + 7$
 $= 3x(x-1) - 7(x-1)$
 $= (3x-7)(x-1)$ } ✓

$y'' = 6x - 10$ ✓

(ii) $y' = 0$ at $x = \frac{7}{3}$ or $x = 1$
 $y = \frac{49}{27}$ $y = 3$
 $y'' = 4$ $y'' = -4$
 > 0 < 0 ✓✓

$(\frac{7}{3}, \frac{49}{27})$ is a minimum turning point } ✓

$(1, 3)$ is a maximum turning point } ✓

(iii) $y'' = 0$
 $6x - 10 = 0$
 $6x = 10$
 $x = \frac{5}{3}$ ✓
 $y = \frac{65}{27}$

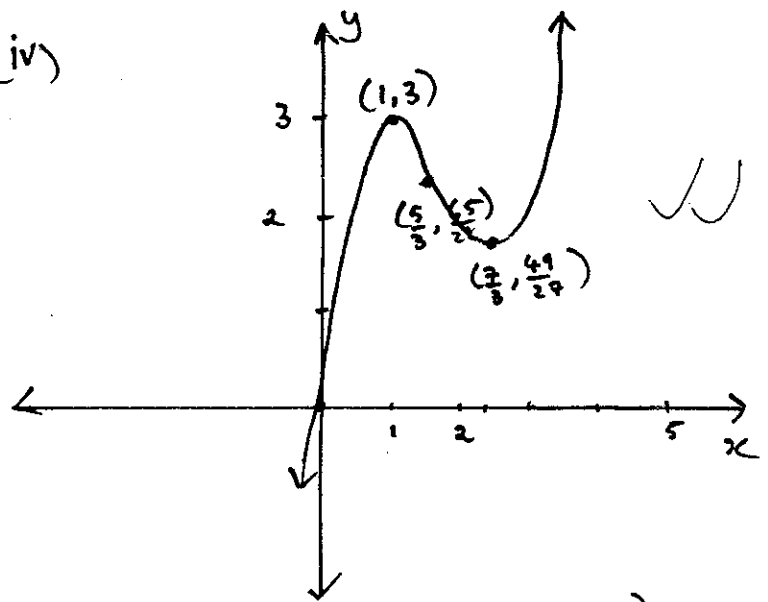
check

x	1	$\frac{5}{3}$	$\frac{7}{3}$
y''	-4	0	4
	n	.	U

 ✓

$(\frac{5}{3}, \frac{65}{27})$ is a point of inflexion

(iv)



(v) check $x = 5$ (end points)
 $y = 5^3 - 5 \times 5^2 + 7 \times 5$
 $= 35$

maximum value is 35 ✓

$$(b) (i) 2x \times 2x \times 2x + 4 \times 2x \times h = 6 \quad \checkmark$$

$$8x^2 + 8xh = 6$$

$$8xh = 6 - 8x^2$$

$$h = \frac{6}{8x} - \frac{8x^2}{8x} \quad \left. \begin{array}{l} \\ \end{array} \right\} \checkmark \text{ either}$$

$$= \frac{3}{4x} - x \quad \text{as required}$$

$$(ii) V = 2x \times 2x \times h$$

$$= 4x^2 \left(\frac{3}{4x} - x \right) \quad \checkmark$$

$$= 3x - 4x^3$$

$$(iii) V' = 3 - 12x^2$$

$$= 3(1 - 4x^2)$$

$$V'' = -24x$$

$$V' = 0 \quad \text{at} \quad 4x^2 = 1$$

$$x^2 = \frac{1}{4}$$

$$x = \frac{1}{2}, \quad x > 0 \quad \checkmark$$

$$V = 3\left(\frac{1}{2}\right) - 4\left(\frac{1}{2}\right)^3 \quad \checkmark$$

$$= 1$$

$$V'' = -24 \quad \checkmark$$

$$< 0$$

the maximum volume is 1 m^3

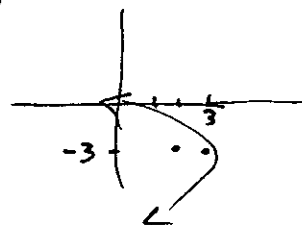


need to
show this

Q5

$$\begin{aligned}
 (a) \quad (i) \quad y^2 + 6y + 4x - 3 &= 0 \\
 y^2 + 6y + 9 &= -4x + 3 + 9 \quad \checkmark \\
 (y+3)^2 &= -4x + 12 \\
 (y+3)^2 &= -4(x-3) \quad \checkmark
 \end{aligned}$$

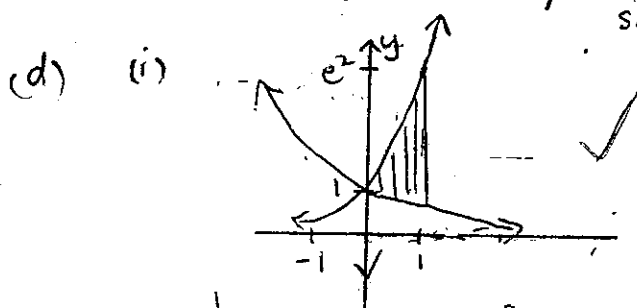
$$\begin{aligned}
 (ii) \quad \text{vertex} &(3, -3) \quad \checkmark \\
 \text{focus} &(2, -3) \quad \checkmark
 \end{aligned}$$



$$\begin{aligned}
 (b) \quad \int_0^3 (2 - \frac{1}{3}x)^{-3} dx &= \left[\frac{(2 - \frac{1}{3}x)^{-2}}{-2 \times -\frac{1}{3}} \right]_0^3 \quad \checkmark \checkmark \\
 &= \frac{3}{2} ((2-1)^{-2} - 2^{-2}) \\
 &= \frac{9}{8} \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 (c) \quad \therefore A &= 2 \int_0^1 e^{-x} dx \quad \checkmark \\
 &= 2 [-e^{-x}]_0^1 \\
 &= 2 (-\frac{1}{e} - (-1)) = 2(1 - \frac{1}{e}) \quad \checkmark
 \end{aligned}$$

square units



$$\begin{aligned}
 V &= \pi \int_{-1}^1 (e^{2x})^2 - (e^{-x})^2 dx \quad \checkmark \\
 &= \pi \int_{-1}^1 e^{4x} - e^{-2x} dx \\
 &= \pi \left[\frac{e^{4x}}{4} + \frac{e^{-2x}}{2} \right]_{-1}^1 \quad \checkmark \\
 &= \pi \left(\frac{e^4}{4} + \frac{1}{2e^2} - \left(\frac{1}{4} + \frac{1}{2} \right) \right) \\
 &= \frac{\pi}{4} (e^4 + \frac{2}{e^2} - 3) \text{ cubic units} \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 (e) \quad \int_k^0 \frac{1}{e^x} dx &= \int_k^0 e^{-x} dx \\
 &= [-e^{-x}]_k^0 \quad \checkmark \\
 &= -1 + e^{-k} \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 -1 + e^{-k} &= e^2 - 1 \\
 e^{-k} &= e^2 \\
 -k &= 2 \\
 k &= -2 \quad \checkmark
 \end{aligned}$$

16

Q6

(a)

(i)

x	-1	0	1
f(x)	$\frac{1}{1+e}$	$\frac{1}{1+1} = \frac{1}{2}$	$\frac{1}{1+e^{-1}}$

(ii) $I \div \frac{1}{2} \left(\frac{1}{1+e} + 2 \times \frac{1}{2} + \frac{1}{1+e^{-1}} \right)$

$$= \frac{1}{2} \left(\frac{1}{1+e} + 1 + \frac{1}{1+\frac{1}{e}} \right)$$

$$= \frac{1}{2} \left(\frac{1}{1+e} + \frac{1+e}{1+e} + \frac{1}{\frac{e+1}{e}} \right)$$

$$= \frac{1}{2} \left(\frac{1}{1+e} + \frac{1+e}{1+e} + \frac{e}{1+e} \right)$$

$$= \frac{1}{2} \left(\frac{2+2e}{1+e} \right)$$

$$= \frac{1}{2} \times 2 \left(\frac{1+e}{1+e} \right)$$

$$= 1$$

(b) $y = x^2 - 2$

(i) $y' = 2x$

at $x = a$

$$y' = 2a$$

normal has equation

$$y - b = -\frac{1}{2a}(x - a)$$

(ii) if $(0,0)$ is on the normal

$$0 - b = -\frac{1}{2a}(0 - a)$$

$$-b = \frac{1}{2}$$

$$b = -\frac{1}{2}$$

$$-\frac{1}{2} = x^2 - 2$$

$$x^2 = \frac{3}{2}$$

$$x = \sqrt{\frac{3}{2}} \text{ or } -\sqrt{\frac{3}{2}}$$

P is $(\sqrt{\frac{3}{2}}, -\frac{1}{2})$ or

$(-\sqrt{\frac{3}{2}}, -\frac{1}{2})$

$$(c) (i) \int_1^5 \frac{1}{x^2} dx = \left[\frac{x^{-1}}{-1} \right]_1^5$$

$$= -\frac{1}{5} - (-1)$$

$$= \frac{4}{5}$$

(ii) the area of the four rectangles is given by

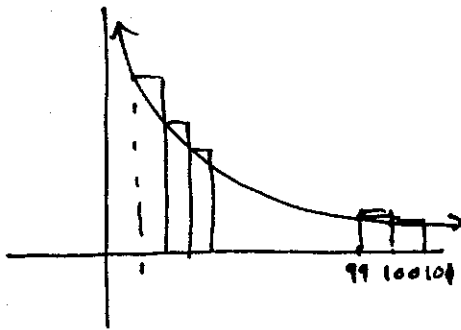
$$1 \times \frac{1}{1^2} + 1 \times \frac{1}{2^2} + 1 \times \frac{1}{3^2} + 1 \times \frac{1}{4^2}$$

the area under the curve is less than the area of the rectangles as parts are above the curve

the area under the curve is $\frac{4}{5}$ from (i)

$$\text{so } 1 \times \frac{1}{1^2} + 1 \times \frac{1}{2^2} + 1 \times \frac{1}{3^2} + 1 \times \frac{1}{4^2} > \frac{4}{5}$$

(iii) extend the diagram to $x=101$



the area under the curve is given by

$$\int_1^{101} \frac{1}{x^2} dx = \left[-x^{-1} \right]_1^{101}$$

$$= -\frac{1}{101} + 1$$

$$= \frac{100}{101}$$

the area of the rectangles is

$$1 \times \frac{1}{1^2} + 1 \times \frac{1}{2^2} + \dots + 1 \times \frac{1}{100^2}$$

using the same reasoning in (ii)

$$\frac{1}{1^2} + \frac{1}{2^2} + \dots + \frac{1}{100^2} > \frac{100}{101}$$

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{99^2} + \frac{1}{100^2} > \frac{99}{100} + \frac{1}{100^2}$$

$$1 + \frac{1}{4} + \frac{1}{9} + \dots + \frac{1}{10000} > \frac{9901}{10000}$$

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